

Operations Research

How to Manipulate Vectors, Index Sets and Matrices

A methods guide for the notation of Lectures 3 to 5: coordinates, x_B and x_N , products, inverses and inequalities

0. How to use this guide

The proofs of the simplex method and of sensitivity analysis are short, but they are written in a dense notation: x_B , A_N , $B^{-1}A_j$, $c_B^\top B^{-1}$, $\sum_{j \in N} r_j x_j$, $y^\top Ax$. Every step in those proofs is one of a handful of moves. This guide lists the moves, says why each one is allowed, and checks each one on numbers.

One example runs through the whole guide. It is the Lecture 5 problem in equational form:

$$\max 3x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 + x_3 = 4, \quad 2x_1 + x_2 + x_4 = 6, \quad x \geq 0,$$

where x_3, x_4 are the slacks s_1, s_2 . So

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \quad x^* = \begin{bmatrix} 2 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

All numbers in this guide were checked by computer.

1. The objects and their sizes

Rule 1: vectors are columns, and sizes must match

- A vector $x \in \mathbb{R}^n$ is a **column** with coordinates $x_1, \dots, x_n$. Its transpose $x^\top$ is the same numbers written as a row.
- A matrix $A \in \mathbb{R}^{m \times n}$ has m rows and n columns. Its entry in row i and column j is a_{ij} . Its j -th **column** is $A_j \in \mathbb{R}^m$, and its i -th **row** is written $a_i^\top$.
- A product PQ is defined only when the number of **columns of P** equals the number of **rows of Q** . Then $(p \times q)(q \times r) = (p \times r)$: the inner sizes must agree and cancel, and the outer sizes give the result.

Checking sizes is the fastest way to catch an error. Every object of the simplex chapters, with its size:

Object	Size	Meaning
A	$m \times n$	constraint matrix
$B = A_B$	$m \times m$	basis matrix (the basic columns)
A_N	$m \times (n - m)$	the nonbasic columns
b	$m \times 1$	right-hand side
x, c, r, d	$n \times 1$	one entry per variable
$x_B, c_B, p, B^{-1}A_j, d_B$	$m \times 1$	one entry per row, or per basic variable
x_N, c_N, r_N	$(n - m) \times 1$	one entry per nonbasic variable
$p^\top = c_B^\top B^{-1}$	$1 \times m$	a row
$c^\top x, p^\top b, r_j, y^\top Ax$	1×1	numbers

Example: a size check catches a wrong formula

Is “ $B^{-1}c_B$ ” the multiplier vector? Its size is $(m \times m)(m \times 1) = m \times 1$, so it is defined, but it is the *wrong* object: the multipliers are $p^\top = c_B^\top B^{-1}$, which is $(1 \times m)(m \times m)$. As a column, $p = (B^{-1})^\top c_B$, which differs from $B^{-1}c_B$ unless B^{-1} is symmetric. In our example $B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$ and $c_B = (3, 2)$:

$$p = (B^{-1})^\top c_B = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{but} \quad B^{-1}c_B = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

2. Coordinates and the unit vectors

The **unit vector** e_i has a 1 in position i and 0 elsewhere. It extracts coordinates, columns and rows:

Rule 2: what e_i picks out

$$e_i^\top x = x_i, \quad Ae_j = A_j \text{ (column } j), \quad e_i^\top A = a_i^\top \text{ (row } i), \quad e_i^\top Ae_j = a_{ij}, \quad x = \sum_{i=1}^n x_i e_i.$$

For a square invertible B : $B^{-1}e_i$ is the i -th *column* of B^{-1} , and $e_i^\top B^{-1}$ is its i -th *row*.

Why

$Ae_j = \sum_k A_k (e_j)_k$, and only the term $k = j$ survives. The same computation on rows gives $e_i^\top A$. The size of e_i is fixed by context: in Ae_j , $e_j \in \mathbb{R}^n$; in $B^{-1}e_i$, $e_i \in \mathbb{R}^m$.

3. Four ways to read a product

(a) **Inner product.** For $u, v \in \mathbb{R}^n$, $u^\top v = \sum_{i=1}^n u_i v_i$ is a *number*, and $u^\top v = v^\top u$. Example: $c^\top x^* = 3 \cdot 2 + 2 \cdot 2 + 0 + 0 = 10$.

(b) **Matrix times vector, Ax .** Two readings, and you need both.

Rule 3: Ax by rows and by columns

$$\begin{aligned} \text{Row view (one equation per row):} \quad & (Ax)_i = \sum_{j=1}^n a_{ij} x_j = a_i^\top x, \\ \text{Column view (a combination of columns):} \quad & Ax = \sum_{j=1}^n x_j A_j. \end{aligned}$$

The row view reads the constraints one at a time. The column view says what $Ax = b$ means: b is built from the columns of A , with weights x_j . The splitting identities of Section 5 come from the column view.

Example: both views of Ax^*

Row view: $(Ax^*)_1 = 1 \cdot 2 + 1 \cdot 2 + 1 \cdot 0 + 0 \cdot 0 = 4$ and $(Ax^*)_2 = 2 \cdot 2 + 1 \cdot 2 = 6$. Column view:
 $Ax^* = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} = b.$

(c) **Row vector times matrix**, $y^\top A$. This is a row of length n , with $(y^\top A)_j = y^\top A_j = \sum_{i=1}^m y_i a_{ij}$: each entry reads one *column* of A down, weighted by y . Transposing, $(y^\top A)^\top = A^\top y$. This is why the dual constraints are $A^\top y \geq c$: dual constraint j is column j of A dotted with y .

(d) **The number $y^\top Ax$** . It can be computed in two orders, and the two orders are the heart of weak duality:

$$y^\top Ax = y^\top (Ax) = (y^\top A) x = \sum_{i=1}^m \sum_{j=1}^n y_i a_{ij} x_j.$$

Example: $y^\top Ax$ two ways, with $y = (1, 1)$

$y^\top (Ax^*) = (1, 1) \cdot (4, 6) = 10$, and $(y^\top A)x^* = (3, 2, 1, 1) \cdot (2, 2, 0, 0) = 6 + 4 = 10$.

(e) **Matrix times matrix**. PQ is the matrix whose j -th column is P times the j -th column of Q : $(PQ)_j = PQ_j$. In particular $B^{-1}A$ has columns $B^{-1}A_j$, and $PQ \neq QP$ **in general**, so never swap the order of a product.

4. Transposes

Rule 4: transpose rules

$$\begin{aligned} (A^\top)^\top &= A, & (PQ)^\top &= Q^\top P^\top, & (Ax)^\top &= x^\top A^\top, \\ (B^{-1})^\top &= (B^\top)^{-1}, & \alpha^\top &= \alpha & \text{for a number } \alpha. \end{aligned}$$

Why

$(PQ)^\top$: entry (i, j) of $(PQ)^\top$ is entry (j, i) of PQ , which is $\sum_k p_{jk} q_{ki} = \sum_k (Q^\top)_{ik} (P^\top)_{kj}$, entry (i, j) of $Q^\top P^\top$. For the inverse, transpose $BB^{-1} = I$ to get $(B^{-1})^\top B^\top = I$. A number is a 1×1 matrix, so it equals its transpose. That is why $y^\top Ax = (y^\top Ax)^\top = x^\top A^\top y$.

5. Index sets: x_B, x_N, A_B, A_N

Rule 5: restricting to an index set

For $S \subseteq \{1, \dots, n\}$:

- x_S is the vector of the coordinates x_j with $j \in S$, and c_S likewise;
- A_S is the matrix made of the *columns* A_j with $j \in S$, in the same order. It keeps all m rows.

A basis is an *ordered* list $B(1), \dots, B(m)$, and $B = A_B = [A_{B(1)} \cdots A_{B(m)}]$. So the k -th entry of x_B is $x_{B(k)}$: **position k in the basis, variable number $B(k)$ in x** . N is the list of the remaining indices.

Example: every piece of the example, with basis (x_1, x_2)

$B(1) = 1$, $B(2) = 2$, and $N = (3, 4)$. Then

$$B = A_B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad A_N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad x_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad x_N = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}, \quad c_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad c_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Row k of the tableau belongs to the variable $x_{B(k)}$: row 1 is $x_1 = \dots$ and row 2 is $x_2 = \dots$

Rule 6: the splitting identities

$$Ax = A_B x_B + A_N x_N, \quad c^\top x = c_B^\top x_B + c_N^\top x_N, \quad u^\top v = u_B^\top v_B + u_N^\top v_N.$$

Why

By the column view, $Ax = \sum_{j=1}^n x_j A_j$. Every index j is either in B or in N , never both, so the sum splits into $\sum_{j \in B} x_j A_j + \sum_{j \in N} x_j A_j$. These two pieces are $A_B x_B$ and $A_N x_N$, again by the column view. The inner products split the same way.

Example: the order of the basis does not matter

List the basis as (x_2, x_1) instead. Then $B' = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ (columns swapped), $c_{B'} = (2, 3)$ and $x_{B'} = (x_2, x_1)$. The tableau rows come out in the other order, but $p^\top = c_{B'}^\top B'^{-1} = (1, 1)$ is unchanged, and so is every reduced cost. In general, reordering is $B' = B\Pi$ for a permutation matrix Π , with $c_{B'} = \Pi^\top c_B$. Then $c_{B'}^\top B'^{-1} = c_B^\top \Pi \Pi^{-1} B^{-1} = c_B^\top B^{-1}$.

Building a vector of $\mathbb{R}^n$ from pieces. The phrase “ $d_j = 1$, $d_i = 0$ for $i \in N \setminus \{j\}$ ”, and $d_B = -B^{-1}A_j$ ” defines all n coordinates of d : one, the others in N , and the m basic ones. In the example, with $j = 3$, this gives $d_B = -B^{-1}A_3 = -B^{-1}e_1 = (1, -2)$, so $d = (1, -2, 1, 0)$. Check: $Ad = (1 - 2 + 1, 2 - 2) = (0, 0)$.

6. Sums with indices, and how to turn them into products

Sum	Product form	Read it as
$\sum_j a_{ij} x_j$	$(Ax)_i = a_i^\top x$	row i of A applied to x
$\sum_i y_i a_{ij}$	$(A^\top y)_j = y^\top A_j$	column j of A weighted by y
$\sum_j x_j A_j$	Ax	combination of columns
$\sum_{j \in N} r_j x_j$	$r_N^\top x_N$	nonbasic part of an inner product
$\sum_{k=1}^m A_{B(k)} d_{B(k)}$	$B d_B$	combination of basic columns
$\sum_{k=1}^m c_{B(k)} x_{B(k)}$	$c_B^\top x_B$	basic part of the objective
$\sum_i \sum_j y_i a_{ij} x_j$	$y^\top Ax$	either order of summation

Rule 7: three habits with indices

- **A summed index is a dummy.** $\sum_j a_{ij} x_j$ and $\sum_k a_{ik} x_k$ are the same thing. Rename freely, but never use one letter for two different roles in the same expression.
- **Finite double sums can be swapped:** $\sum_i \sum_j = \sum_j \sum_i$. This is exactly $y^\top (Ax) = (y^\top A)x$.

- **Constants come out of sums:** $\sum_j \alpha u_j = \alpha \sum_j u_j$, and $\sum_j (u_j + v_j) = \sum_j u_j + \sum_j v_j$.

7. Inverses: what B^{-1} lets you do

Rule 8: the inverse

- B^{-1} exists only for a **square** matrix with independent columns, and $BB^{-1} = B^{-1}B = I$.
- $Bu = v \iff u = B^{-1}v$, by multiplying **on the left** by B^{-1} or by B . This is the only way to “undo” a matrix. There is no division by a matrix.
- $(PQ)^{-1} = Q^{-1}P^{-1}$, in reverse order, like the transpose.
- $B^{-1}A_{B(k)} = e_k$, since $B^{-1}B = I$ column by column. So $B^{-1}A$ has the unit vectors in the basic columns.
- For a 2×2 matrix: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Example: computing without forming B^{-1} first

$$B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \text{ has } ad - bc = 1 - 2 = -1, \text{ so } B^{-1} = - \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}.$$

- $x_B = B^{-1}b = (-4 + 6, 8 - 6) = (2, 2)$. Equivalently, solve $Bx_B = b$: $x_1 + x_2 = 4$, $2x_1 + x_2 = 6$.
- $p^\top = c_B^\top B^{-1}$. Transposing gives $B^\top p = c_B$, that is $p_1 + 2p_2 = 3$ and $p_1 + p_2 = 2$, so $p = (1, 1)$.
- $r_j = c_j - p^\top A_j$: $r_3 = 0 - (1, 1) \cdot (1, 0) = -1$, $r_4 = 0 - (1, 1) \cdot (0, 1) = -1$, and $r_1 = 3 - (1, 1) \cdot (1, 2) = 0$, as a basic reduced cost must be.

8. Inequalities between vectors

Rule 9: componentwise order

$u \leq v$ means $u_i \leq v_i$ for *every* i . You may:

- add two inequalities, or add the same vector to both sides;
- multiply both sides by a number $\alpha \geq 0$;
- multiply by a **nonnegative** matrix: if $u \leq v$ and $M \geq 0$ entrywise, then $Mu \leq Mv$;
- take an inner product with a nonnegative vector: if $u \leq v$ and $w \geq 0$, then $w^\top u \leq w^\top v$.

Why

$w^\top v - w^\top u = \sum_i w_i(v_i - u_i)$, a sum of products of two nonnegative numbers. Each row of M is such a w . This single fact is behind every line of weak duality: $c \leq A^\top y$ and $x \geq 0$ give $c^\top x \leq y^\top Ax$, and $Ax \leq b$ and $y \geq 0$ give $y^\top Ax \leq y^\top b$.

Example: you may *not* multiply an inequality by B^{-1}

$u = (0, 0) \leq v = (1, 0)$. But $B^{-1}u = (0, 0)$ and $B^{-1}v = (-1, 2)$, and $0 \leq -1$ is false. B^{-1} has negative entries, so Rule 9 does not apply. This is why feasibility $B^{-1}b \geq 0$ must be *checked*, never derived from $b \geq 0$. *Equalities* can be multiplied by any matrix.

Two more traps. Vectors cannot always be compared: neither $(1, 2) \leq (2, 1)$ nor $(2, 1) \leq (1, 2)$ holds. And $x \geq 0$ with $x \neq 0$ is much weaker than $x > 0$, which requires every coordinate to be positive.

9. The five moves of the proofs, done slowly

Every proof in Lectures 3 to 5 is a combination of these moves.

Move 1: split, solve, substitute (the reduced costs)

Start from $Ax = b$ and split it (Rule 6): $Bx_B + A_Nx_N = b$. Solve for x_B by multiplying on the left by B^{-1} (Rule 8): $x_B = B^{-1}b - B^{-1}A_Nx_N$. Substitute into the split objective:

$$\begin{aligned} c^\top x &= c_B^\top x_B + c_N^\top x_N = c_B^\top (B^{-1}b - B^{-1}A_Nx_N) + c_N^\top x_N \\ &= c_B^\top B^{-1}b + (c_N^\top - c_B^\top B^{-1}A_N)x_N = c_B^\top B^{-1}b + r_N^\top x_N. \end{aligned}$$

The bracket is a row of length $n - m$, and its entry for $j \in N$ is $c_j - c_B^\top B^{-1}A_j = r_j$. Example: $c^\top x = 10 - x_3 - x_4$, the objective row of the final tableau.

Move 2: one coordinate is free, the equations fix the rest (basic directions)

To find d with $Ad = 0$, $d_j = 1$ and $d_i = 0$ for the other nonbasic i : split $Ad = Bd_B + A_Nd_N$, where $A_Nd_N = A_j$ because $d_N = e_j$ (Rule 2). So $Bd_B + A_j = 0$, and $d_B = -B^{-1}A_j$.

Move 3: insert the middle term (duality)

To compare $c^\top x$ with $b^\top y$, write the number $y^\top Ax$ between them and use one inequality on each side (Rule 9):

$$c^\top x \leq (A^\top y)^\top x = y^\top Ax = y^\top (Ax) \leq y^\top b.$$

The equalities in the middle are Rule 4 and the associativity of products.

Move 4: plug in special vectors to identify coefficients

If an identity $\alpha + \rho^\top x_N = \alpha' + \rho'^\top x_N$ holds for *every* x_N , then putting $x_N = 0$ gives $\alpha = \alpha'$, and putting $x_N = e_j$ gives $\rho_j = \rho'_j$. This is how one proves that the tableau of a basis is unique.

Move 5: prove a vector statement one coordinate at a time

To show $A^\top p \geq c$, show $(A^\top p)_j \geq c_j$ for each j : $(A^\top p)_j = p^\top A_j$, and $r_j = c_j - p^\top A_j \leq 0$ says exactly that. For a slack column $A_j = e_i$ with $c_j = 0$, $r_j = -p^\top e_i = -p_i$, so $r_j \leq 0$ means $p_i \geq 0$.

10. A full computation: the production example

$\max 4x_1 + 3x_2 + 2x_3 \quad \text{s.t.} \quad x_1 + x_2 + x_3 + x_4 = 8, \quad 2x_1 + x_2 + x_3 + x_5 = 10, \quad x \geq 0,$
with basis $B = (1, 2)$ and $N = (3, 4, 5)$.

Object	Computation	Result
B, A_N	columns 1, 2; columns 3, 4, 5	$\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$
B^{-1}	2×2 formula, $\det = -1$	$\begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$
$x_B = B^{-1}b$	$(-8 + 10, 16 - 10)$	$(2, 6) \geq 0$: feasible
$B^{-1}A_N$	B^{-1} times each column of A_N	$\begin{bmatrix} 0 & -1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$
p	solve $B^\top p = c_B$: $p_1 + 2p_2 = 4,$ $p_1 + p_2 = 3$	$p = (2, 1)$
$r_N = c_N - A_N^\top p$	$(2, 0, 0) - (3, 2, 1)$	$(-1, -2, -1) \leq 0$: optimal

The tableau is $x_B = B^{-1}b - B^{-1}A_N x_N$ and $z = c_B^\top x_B + r_N^\top x_N$:

$$x_1 = 2 + 0x_3 + x_4 - x_5, \quad x_2 = 6 - x_3 - 2x_4 + x_5, \quad z = 26 - x_3 - 2x_4 - x_5,$$

exactly the tableau of Lecture 5, slide 45. Note the minus sign: the tableau shows $-B^{-1}A_N$.

11. Exercises

Answers are at the end of the list. Use the running example where needed.

- What is the size of $c_B^\top B^{-1}A_N$? Of $B^{-1}A_N x_N$?
- Compute $e_2^\top A$ and Ae_3 .
- Show that $(y^\top A)_j = y^\top A_j$ using sums.
- Write $\sum_{k=1}^m \sum_{j \in N} c_{B(k)} (B^{-1})_{k \cdot} A_j x_j$ as a product of vectors and matrices.
- Is $x_N^\top A_N^\top$ a row or a column? What is its size?
- Prove $(B^{-1}A)e_j = B^{-1}A_j$.
- True or false: if $b \geq 0$, then $B^{-1}b \geq 0$.
- With $p = (1, 1)$ and the new column $a = (1, 1)$, compute $p^\top a$ and $\gamma - p^\top a$ for $\gamma = 2.5$.

Answers.

- $(1 \times m)(m \times m)(m \times (n - m)) = 1 \times (n - m)$, a row. Then $(m \times m)(m \times (n - m))((n - m) \times 1) = m \times 1$, a column.
- $e_2^\top A = (2, 1, 0, 1)$, the second row. $Ae_3 = (1, 0)$, the third column.
- $(y^\top A)_j = \sum_i y_i a_{ij}$, and the column A_j has entries a_{ij} , so this is $\sum_i y_i (A_j)_i = y^\top A_j$.
- $c_B^\top B^{-1}A_N x_N = p^\top A_N x_N$, a number.
- $x_N^\top A_N^\top = (A_N x_N)^\top$ is a row of size $1 \times m$.
- By the column rule (Section 3e), column j of $B^{-1}A$ is B^{-1} times column j of A , and $Ae_j = A_j$. Formally, $(B^{-1}A)e_j = B^{-1}(Ae_j) = B^{-1}A_j$, by associativity.

- (g) False. B^{-1} can have negative entries (Section 8). Here, with $b = (1, 0)$, $B^{-1}b = (-1, 2)$.
- (h) $p^\top a = 2$, and $2.5 - 2 = +0.5 > 0$: the new activity should enter.

Common mistakes

- **Swapping a product.** $c_B^\top B^{-1} \neq B^{-1}c_B$, and $PQ \neq QP$. Check sizes.
- **Multiplying on the wrong side.** From $Bu = v$, multiply *on the left* by B^{-1} . Writing $u = vB^{-1}$ is meaningless (the sizes do not match).
- **Multiplying an inequality by a matrix with negative entries.** Only nonnegative matrices and vectors preserve $\leq$.
- **Confusing position and variable.** $x_{B(k)}$ is the k -th basic variable, not x_k . In the production example, $B = (1, 2)$ makes them agree by luck. With basis $(2, 1)$, $x_{B(1)} = x_2$.
- **Forgetting that A_S keeps all rows.** A_N selects columns, never rows.
- **Treating $\sum_{j \in N}$ as $\sum_{j=1}^n$.** The basic terms are not in the sum. They are zero anyway when the coefficients are reduced costs, since $r_B = 0$, but not in general.
- **Transposing without reversing.** $(PQ)^\top = Q^\top P^\top$ and $(PQ)^{-1} = Q^{-1}P^{-1}$.